Reusable, cross-species–validated pipeline for tissue-specific AAV regulatory elements

One pipeline · any cell type · only the argument changes — a synthetic-biology “parts registry” for gene-therapy design

WHY IT MATTERS

- Tissue-restricted expression → lower off-target toxicity, dose & immune load
- Cross-species failure is the field's costliest failure mode:

elements tuned in mouse routinely fail in human/NHP

- We flag likely non-translating parts computationally, up front

REUSABILITY IS STRUCTURAL, NOT CLAIMED

```
$ python run.py "Pancreatic_Beta_Cell"
$ python run.py "Small_Intestinal_Enterocyte"
```

Code lines changed between the two runs: **0**

Only the cell-type string differs. No per-tissue branching.

OPEN DATA

CATLAS scATAC (1.14M cCRE × 222 cell types) · JASPAR · GENCODE
hg38 · CELLxGENE Census · Ensembl orthologs | code: MIT

THE FUNNEL (same thresholds, both tissues)

Union cCREs	1,143,424
Accessible	181,897
Specific ($\tau \geq 0.9$)	82,038
Size ≤800 bp	65,101
Ranked shortlist	25
Cross-species conserved	7

~1.1M candidate elements → a ranked, translation-aware shortlist

POSITIVE CONTROL ✓

Top beta parts at INS (rank 1) + IAPP
Enterocyte: OLFM4 / APOA4 / PIGR
Housekeeping (ACTB…) → $\tau \approx 0.02$, filtered

MOTIF ENRICHMENT ✓

Beta shortlist 4.7× enriched
for PDX1/NKX6-1/PAX6/NEUROD1
$p = 1.6\times10^{-13}$ (72% vs 12% PDX1)

CROSS-SPECIES ★ headline

Each part flagged conserved / high-risk
Enterocyte PHGR1 → no_ortholog:
human-specific; a mouse-only run misses it